

1 Basics

- Strings are immutable sequences of characters.
- Can be defined using single, double, or triple quotes.

```
s1 = "Hello"  
s2 = 'World'  
s3 = """Multiline string"""
```

- Indexing:

```
s = "Python"  
print(s[0])    # 'P'  
print(s[-1])   # 'n'
```

- Slicing:

```
print(s[0:4])   # 'Pyth'  
print(s[2:])    # 'thon'  
print(s[::-1])  # 'nohtyP'
```

2 String Methods

- Changing case:

```
s.upper(), s.lower(), s.title(), s.capitalize()
```

- Searching / Checking:

```
s.find('o'), s.count('o'), s.startswith('Py'), s.endswith('on')
```

- Splitting / Joining:

```
s.split(), '-'.join(['Python', 'Rocks'])
```

- Replacing / Stripping:

```
s.replace('Py', 'Ju'), s.strip(), s.lstrip(), s.rstrip()
```

3 String Operations

- Concatenation: `"Hello" + "World"` → `'HelloWorld'`
- Repetition: `"Hi" * 3` → `'HiHiHi'`
- Membership check:

```
'Py' in "Python" # True  
'th' not in "Python" # False
```

4 String Formatting

- f-strings:

```
name = "Ali"  
age = 25  
f"My name is {name} and age is {age}"
```

- `.format()` method:

```
"My name is {} and age is {}".format(name, age)  
"My name is {name} and age is {age}".format(name="Ali", age=25)
```

- Old style `%` formatting:

```
"My name is %s and age is %d" % (name, age)
```

5 Useful Methods & Use-Cases

Method	Returns	Use-case
<code>startswith(sub)</code>	True/False	Validate input / filter lines / check URL prefix
<code>endswith(sub)</code>	True/False	Check file type / filter lines / check suffix
<code>lstrip()</code>	string	Remove leading spaces / unwanted chars
<code>rstrip()</code>	string	Remove trailing spaces / unwanted chars
<code>count()</code>	int	Count occurrences of a character
<code>replace()</code>	string	Replace substring in a string

Method	Returns	Use-case
split()	list	Split string into words or elements
join()	string	Join list elements into string
find()	int	Find index of substring

6 Example Problems

1. Count vowels and consonants in a string

```
text = "Hello World"
vowels = 0
consonants = 0
for char in text:
    if char.isalpha():
        if char.lower() in 'aeiou':
            vowels += 1
        else:
            consonants += 1
print("Vowels:", vowels)
print("Consonants:", consonants)
```

2. Reverse a string
3. Check if string is palindrome
4. Remove spaces / punctuation
5. Count frequency of each character
6. Replace all occurrences of a word
7. Extract substring using slicing
8. Check if string starts with/ends with a substring
9. Convert string to uppercase/lowercase/titlecase

 **Tips for Practice:** - Start with indexing & slicing problems - Then use string methods like split, join, replace - Combine loops + strings for counting & frequency problems - Gradually move to bigger problems: palindrome, word frequency